Wednesday, 03 July 2019

8:30	**Registration**
9:00	**Opening and Welcome**
9:10	**Keynote 1 Professor Veronique Perdereau** **Chair:** Professor Kaspar Althoefer
10:10	**Coffee Break**
	Oral Presentation Session 1-1
	Topic: Robotic Grippers and Manipulation **Chair:** Dr Lorenzo Jamone
10:30	**Reasoning on Grasp-Action Affordances** Paola Ardón*, Èric Artau*, Ron Petrick, Subramanian Ramamoorthy, Katrin Solveig Lohan
10:45	**Design analysis of a fabric based lightweight robotic gripper** Ahmed Hassan*, Hareesh Godaba*, Kaspar Althoefer
11:00	**A method to estimate the oblique arch folding axis for thumb assistive devices** *Visakha Nanayakkara*, Nantachai Sornkaran, Hasitha Wegiriya, Nikolaos Vitzilaios, Demetrios Venetsanos, Nicolas Rojas, M. Necip Sahinkaya, Thrishantha Nanayakkara*
11:15	**Energy-Tank based Force Control for 3D Contour Following** *Salua Hamaza*, Ioannis Georgilas, Tom Richardson*
11:30	**Kinematic Control and Obstacle Avoidance for Soft Inflatable Manipulator** *Ahmad Ataka*, Agostino Stilli, Jelizaveta Konstantinova, Helge Wurdemann, Kaspar Althoefer*
11:45	**Learning and Composing Primitive Skills for Dual-arm Manipulation** *Èric Artau*, Paola Ardón*, Michael Mistry*, Yvan Petillot**
12:00	**DE VITO: A Dual-arm, High Degree-of-freedom, Lightweight, Inexpensive, Passive Upper-limb Exoskeleton for Robot Teleoperation** *Fabian Falck*, Kawin Larppichet, Petar Kormushev*
12:15	**A Novel Probabilistic Projection Model for Multi-Camera Object Tracking** *Jiaxin Lin*, Chun Xiao, Disi Chen, Dalin Zhou, Zhaojie Ju, Honghai Liu*
12:30	**Lunch and exhibition**
	Oral Presentation Session 1-2
	Topic: Soft Robotics, Sensing and Mobile Robots **Chair:** Dr Myra Wilson
13:30	**Soft fiber-reinforced pneumatic actuator design and fabrication: Towards robust, soft robotic systems** *Jan Fras*, Kaspar Althoefer*
13:45	**Ultrasound Feature Evaluation for Robustness to Sensor Shift in Ultrasound Sensor Based Hand Motion Recognition** *Peter Boyd, Yinfeng Fang, Honghai Liu*
14:00	**Light Intensity-Modulated Bending Sensor Fabrication and Performance Test for Shape Sensing** *Faisal ALJaber*, Kaspar Althoefer**
14:15	**Designing Origami-adapted Deployable Modules for Soft Continuum Arms** *Ketao Zhang*, Kaspar Althoefer**
14:30	**A Debris Clearance Robot for Extreme Environments** Craig West*, Farshad Arvin*, Wei Cheah, Andrew West, Simon Watson, Manuel Giuliani, Barry Lennox
14:45	**Dynamic Response Characteristics in Variable Stiffness Soft Inflatable Links** *Ahmad Ali, Kaspar Althoefer, Jelizaveta Konstantinova*
15:00	**Investigating balance control of a hopping bipedal robot** *Beichen Ding*, Andrew Plummer, Pejman Iravani*
15:15	**Continuous Motion Utilising Advanced Motions on a Hexapod** *Wei Chen Cheah, Hassan Hakim Khalili, Simon Watson, Peter Green, Barry Lennox*
15:30	**Invited talk - Rich Walker**
16:00	**Coffee Break**

		Poster Session 1
16:00 - 17:00	P1	**Making the Case for Human-aware Navigation in Warehouses** Manuel Fernandez-Carmona*, Tejas Parekh, Marc Hanheide
	P2	**A vision-based assistance key differenciator for helicopters automonous scalable missions** Rémi Girard*, Sébastien Mavromatis, Jean Sequeira, Nicolas Belanger, Guillaume Anoufa
	P3	**The Third hand, Cobots Assisted Precise Assembly** Mohammad Safeea, Pedro Neto*, Richard Bearee
	P4	**Towards a Swarm Robotic System for Autonomous Cereal Harvesting** Alan Millard*, Roopika Ravikanna*, Roderich Gross*, David Chesmore*
	P5	**Optimal Manoeuver Trajectory Synthesis for Autonomous Space and Aerial Vehicles and Robots** Ranjan Vepa*
	P6	**Soft particles for granular jamming** Fabrizio Putzu*, Jelizaveta Konstantinova*, Kaspar Althoefer
	P7	**Intuitive Bare-Hand Teleoperation of a Robotic Manipulator using Virtual Reality and Leap Motion** Inmo Jang*, Joaquin Carrasco Gomez, Andrew Weightman, Barry Lennox
	P8	**An Optimal Approach to Anytime Task and Path Planning for Autonomous Mobile Robots in Dynamic Environments** Cuebong Wong*, Erfu Yang, Xiu-Tian Yan, Dongbing Gu
	P9	**A Self-Organizing Network with Varying Density Structure for Characterizing Sensorimotor Transformations in Robotic Systems** Omar Zahra*, David Navarro-Alarcon*
	P10	**A Cross-Landscape Evaluation of Multi-Robot Team Performance in Static Task-Allocation Domains** Dingdian Zhang*, Eric Schneider, Elizabeth Sklar*
	P11	**Payload Capabilities and Operational Limits of Eversion Robots** Hareesh Godaba*, Fabrizio Putzu, Taqi Abrar, Jelizaveta Konstantinova, Kaspar Althoefer
	P12	**Random walk exploration for swarm mapping** Miquel Kegeleirs*, David Garzón Ramos*, Mauro Birattari*
	P13	**MoDSeM: Towards Semantic Mapping with Distributed Robots** Gonçalo Martins*, João Filipe Ferreira, David Portugal, Micael Couceiro
	P14	**Boundary Detection in a Swarm of Kilobots** Yingyi Kuang, Yuri Kaszubowski Lopes*, Roderich Gross*
	P15	**Robot path planning using imprecise and sporadic advisory information from humans** Gianni Di Caro*, Eduardo Feo*
	P16	**A K-Nearest Neighbours Based Inverse Sensor Model for Occupancy Mapping** Yu Miao, Ioannis Georgilas, Alan Hunter
17:00		**Welcome Reception**
18:00	**IET Public Talk, Professor Aude Billard** **Chair: Professor Kaspar Althoefer**	

Thursday, 04 July 2019

09:00	**Keynote 2 Dr Francesco Nori** **Chair: Professor Stefan Edelkamp**
10:00	**Coffee Break**
	Oral Presentation Session 2-1
	Topic: **Robotic Learning, Mapping and Planning** **Chair:** **Dr Alessandro Di Nuovo**
10:20	**Towards adversarial training for mobile robots** *Todd Flyr*, Simon Parsons**
10:35	**Collaborative HRI and Machine Learning for Constructing Personalised Physical Exercise Databases** *Daniel Delgado Bellamy*, Praminda Caleb-Solly**
10:50	**ORB-SLAM-CNN: Lessons in Adding Semantic Map Construction to Feature-Based SLAM** Andrew Webb*, Gavin Brown, Mikel Lujan
11:05	**Probabilistic Planning for Robotics with ROSPlan** *Gerard Canal*, Michael Cashmore, Senka Krivic, Guillem Alenyà, Daniele Magazzeni, Carme Torras*
11:20	**Coverage Path Planning for Large-Scale Aerial Mapping** *Nasser Gyagenda*, Ahmad Kamal Nasir, Hubert Roth, Vadim Zhmud*
11:35	**Self-organized Collective Motion with a Simulated Real Robot Swarm** *Mohsen Raoufi*, Ali Emre Turgut, Farshad Arvin**
11:50	**Contact Planning for the ANYmal Quadruped Robot using an Acyclic Reachability-Based Planner** *Mathieu Geisert*, Thomas Yates, Asil Orgen, Pierre Fernbach, Ioannis Havoutis*
12:05	**Multi-Robot Multi-Goal Motion Planning with Time and Resources** *Stefan Edelkamp**
12:30	**Lunch and exhibition**
	Oral Presentation Session 2-2
	Topic: **Human-Robot Interaction** **Chair:** **Dr Alan Millard**
13:30	**instruMentor: an Interactive Robot for Musical Instrument Tutoring** *Shreyus Bagga, Benedikt Maurer, Tom Miller, Luke Quinlan, Lorenzo Silvestri, Dan Wells, Rebecka Winqvist, Mark Zolotas*, Yiannis Demiris*
13:45	**Position and Velocity Control for Telemanipulation with Interoperability Protocol** *Bukeikhan Omarali*, Francesca Palermo, Maurizio Valle, Stefan poslad, Kaspar Althoefer, Ildar Farkhatdinov**
14:00	**Intrinsically Motivated Autonomy in Human-Robot Interaction: Human Perception of Predictive Information in Robots** *Marcus M. Scheunemann*, Christoph Salge, Kerstin Dautenhahn*
14:15	**Modeling and Control of Ankle Actuation Platform for Human-Robot Interaction** *Ata Otaran*, Ildar Farkhatdinov**
14:30	**Investigating the effects of social interactive behaviours of a robot on people's trust during a navigation task** *Alessandra Rossi*, Fernando Garcia, Arturo Cruz-Maya, Kerstin Dautenhahn, Kheng Lee Koay, Michael Walters, Amit K. Pandey*
14:45	**A Dataset for Action Recognition in the Wild** Alexander Gabriel*, Serhan Cosar, Nicola Bellotto, Paul Baxter*
15:00	**Feel it on your Fingers: Dataglove with Vibrotactile Feedback for Virtual Reality and Telerobotics** Burathat Junput, Xuyi Wei, Lorenzo Jamone*
15:15	**Virtual Reality Simulator for Medical Auscultation Training** *Luis Andres Puertolas Balint*, Luis Humberto Perez Macías, Kaspar Althoefer*
15:30	**QMUL UK Best PhD in Robotics Award Session**
16:00	**Coffee Break**

		Poster Session 2
16:00 - 17:00	P17	**Mobile Robot Trajectory Analysis with the Help of Vision System** *Dinmohamed Danabek*, Ata Otaran, Ildar Farkhatdinov**
	P18	**Autonomous Air-Hockey Playing Cobot using Optimal Control and Vision-Based Bayesian Tracking** Ahmad AlAttar, Louis Rouillard, Petar Kormushev
	P19	**LibRob: an Autonomous Assistive Librarian** Costanza Di Veroli, Cao An Le, Thibaud Lemaire, Eliot Makabu, Abdullahi Nur, Vincent Ooi, Jee Yong Park, Federico Sanna, Rodrigo Chacon Quesada, Yiannis Demiris
	P20	**Watchman Routes for Robot Inspection** Stefan Edelkamp*
	P21	**Visual and Thermal Data for Pedestrian and Cyclist Detection** Sarfraz Ahmed*, M. Nazmul Huda*, Sujan Rajbhandari*, Chitta Saha*, Stratis Kanarachos*, Mark Elshaw*
	P22	**An Augmented Reality Environment to Provide Visual Feedback to Amputees during sEMG Data Acquisitions** Francesca Palermo*, Matteo Cognolato, Ivan Eggel, Manfredo Atzori, Henning Müller
	P23	**Online Human In-Hand Manipulation Skill Recognition and Learning** Disi Chen, Zhaojie Ju*, Dalin Zhou, Gongfa Li, Honghai Liu
	P24	**Towards Generating Simulated Walking Motion using Position Based Deep Reinforcement Learning** William Jones, Siddhant Gangapurwala, Ioannis Havoutis, Kazuya Yoshida
	P25	**Modelling of a soft sensor for exteroception and proprioception in a pneumatically actuated soft robot** Abu Bakar Dawood*, Hareesh Godaba*, Kaspar Althoefer*
	P26	**A Robust Polyurethane Depositing System for Overcoming Obstacles in Disaster Scenario Robotics** Alec Burns*. Sebastiano Fichera. Paolo Paoletti
	P27	**Improved Safety with a Distributed Routing Strategy for UAVs** William Bonnell*
	P28	**Robotic-assisted Ultrasound for Fetal Imaging: Evolution from Single-arm to Dual-arm System** Shuangyi Wang*, James Housden, Yohan Noh, Davinder Singh, Anisha Singh, Emily Skelton, Jacqueline Matthew, Cornelius Tan, Junghwan Back, Lukas Lindenroth, Alberto Gomez, Nicolas Toussaint, Veronika Zimmer, Caroline Knight, Tara Fletcher, David Lloyd, John Simpson, Dharmintra Pasupathy, Hongbin Liu, Kaspar Althoefer, Joseph Hajnal, Reza Razavi, Kawal Rhode
	P29	**Evaluating ToRCH structure for characterizing robots** Manal Linjawi*, Roger Moore
	P30	**Towards Long-Term Autonomy based on Temporal Planning** Yaniel Carreno*, Ron Petrick, Yvan Petillot
	P31	**A Quest Towards Safe Human Robot Collaboration** Mohammad Safeea, Pedro Neto*, Richard Bearee
	P32	**MRComm: Multi-Robot Communication Testbed** Tsvetan Zhivkov*, Eric Schneider, Elizabeth Sklar*
19:00		**Dinner**

Friday, 05 July 2019

09:00		**Keynote 3 Professor Bruno Siciliano** **Chair: Dr Ketao Zhang**
10:00		**Coffee Break**
		Poster Session 3
10:00 - 11:00	P33	**Omni-Pi-tent: An Omnidirectional Modular Robot with Genderless Docking** Robert Peck, Jonathan Timmis, Andy Tyrrell
	P34	**Exploration: Do we need a map?** Mohamed Idries, Matthias Rolf, Tjeerd olde Scheper
	P35	**System Design and Control of a Di-Wheel Rover** John Koleosho
	P36	**Eduardo: A Low Cost Assistive Robot Development Platform Featuring a Compliant End Effector** Oliver Smith*, Samuel White*
	P37	**GarmNet: Improving Global with Local Perception for Robotic Laundry Folding** Daniel Gomes*, Shan Luo*, Luis Teixeira*
	P38	**Wheelchair Navigation: Automatically Adapting to Evolving Environments** Tomos Fearn*, Frederic Labrosse, Patricia Shaw
	P39	**An Auto-Correction Teleoperation Method for a Mobile Manipulator Using Gaze Tracking and Hand Motion Detection** Junshen Chen*, Ze Ji*, Hanlin Niu*, Rossi Setchi*, Chenguang Yang*
	P40	**Collision-Free Optimal Trajectory for a Controlled Floating Space Robot** Asma Seddaoui*, Chakravarthini Saaj
	P41	**Development and Evaluation of a Novel Robotic System for Search and Rescue** Andrea Cachia, M. Nazmul Huda*, Pengcheng Liu, Chitta Saha, Andrew Tickle, John Arvanitakis, Syed Mahfuzul Aziz
	P42	**The Impact of the Robot's Morphology in the Collective Transport** Jessica Meyer*
	P43	**Mapping of Ultra-Wide Band Positional Variance for Indoor Environments** Harry Pointon*, Frederic Bezombes
	P44	**The Downsizing of a Free-Flying Space Robot** Lucy Jackson*, Chakravarthini Saaj, Asma Seddaoui, Steve Eckersley, Calem Whiting
	P45	**Semantic Path Planning for Indoor Navigation and Household Tasks** Nico Sun*, Erfu Yang, Jonathan Corney, Leo Chen
	P46	**Preliminary investigation on visual finger-counting with the iCub robot cameras and hands** Alexandr Lucas*, Carlos Ricolfe-Viala*, ALESSANDRO DI NUOVO*
	P47	**Elastomer-based Touch Sensor: Visualization of Tactile Pressure Distribution** Wanlin Li, Jelizaveta Konstantinova, Akram Alomainy, Kaspar Althoefer*
	P48	**Modular, Underactuated Anthropomorphic Robot Hand with Flexible Fingers and Twisted String Actuators** Faiz Rahman, Kaiqiang Zhang, Guido Herrmann

	Oral Presentation Session 3	
	Topic:	**Robotic Systems and Applications**
	Chair:	**Prof Pedro Neto**
11:00	**Note on Geometric and Exponential Expressions of Screw Displacement** *Guowu Wei*, Anthony H. Jones, Lei Ren*	
11:15	**A modular 3D-printed inverted pendulum** *Ian Howard**	
11:30	**A Geometric Dynamics Algorithm for Serially Linked Robots** *Mohammad Safeea, Pedro Neto*, Richard Bearee*	
11:45	**Full-Rotation Singularity-Safe Workspace for Kinematically Redundant Parallel Robots** *Nicholas Baron*, Yat Hei Cheung*, Nicolas Rojas**	
12:00	**Model-based 3D point cloud segmentation for automated selective broccoli harvesting** *Hector A. Montes*, Grzegorz Cielniak, Tom Duckett*	
12:15	**Mine Detonating Sphere-Bot** *Rebecca Harding*, Charlie Freestone*, Martin Stoelen**	
12:30	**Invited talk - Dr Stoyan Smoukov** **BOTTOM-UP ROBOTICS – TOWARDS GROWING MATERIAL ROBOTS**	
13:00	**Awards Ceremony**	
13:30	**Lunch**	